

IPO SUCCESS DRIVERS IN INDIA (2010–2021)

A Data-Driven Analysis of Factors Influencing IPO Listing Performance

Author: Gopal Agarwal

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Tools Used

Python, Pandas, NumPy, SciPy, Statsmodels, Excel, Power BI

1. Abstract

Initial Public Offerings (IPOs) attract significant investor attention due to their potential for generating listing-day gains. However, IPO performance varies considerably across companies and market conditions. This study analyzes 264 Indian IPOs issued between 2010 and 2021 to identify the major factors influencing listing-day returns.

The research evaluates the impact of subscription demand, institutional participation, issue size, and market sentiment on IPO performance using statistical analysis, hypothesis testing, and regression modeling.

2. Objectives

The objectives of this study are:

To identify the key drivers of IPO listing gains.

To analyze the role of investor demand.

To evaluate the effect of institutional participation.

To study the impact of issue size.

To examine the influence of market sentiment.

3. Dataset Description

The study utilizes two datasets to analyze the determinants of IPO listing performance in India.

3.1 IPO Dataset

The primary dataset consists of 264 Indian IPOs listed between 2010 and 2021. The data was compiled from publicly available IPO databases and market sources.

The dataset contains the following variables:

IPO Name

Listing Date

Company Profile

Issue Size (■ Crores)

QIB Subscription

HNI Subscription

Retail Subscription

Total Subscription

Issue Price

Listing Open Price

Listing Close Price

Listing Gains (%)

Engineered Variables

Several additional variables were created during the analysis:

Outcome Category

Disaster
Loss Making
Moderate Success
Strong Success
Blockbuster Success
Subscription Category
High Demand
Medium Demand
Low Demand
Under Subscribed
Institutional Confidence
Very High Institutional Interest
High Institutional Interest
Moderate Institutional Interest
Low Institutional Interest
Subscription Mismatch
Institution Driven
Retail Driven
Issue Size Category
Small
Medium
Large

3.2 Market Sentiment Dataset

To capture overall market conditions, historical NIFTY 50 index data was collected from the National Stock Exchange covering the period from 2010 to 2021.

The NIFTY dataset contains:

Date
Open Price
High Price
Low Price
Close Price
Shares Traded
Turnover

Using the NIFTY closing prices, market sentiment indicators were constructed.

Derived Variables
NIFTY 30-Day Return
NIFTY 90-Day Return

The 90-day return was used as a proxy for market sentiment and was merged with the IPO dataset based on listing dates.

Final Dataset

After feature engineering and merging the market data, the final analytical dataset consisted of:

264 IPO observations

20+ variables

Financial variables

Demand variables

Institutional participation variables

Market sentiment indicators

This integrated dataset enabled a comprehensive analysis of both company-level and market-level factors influencing IPO listing performance.

4. Methodology

The analysis was conducted in five stages:

Data Cleaning and Preparation

Exploratory Data Analysis

Feature Engineering

Statistical Testing

Regression Modeling

Techniques used:

Correlation Analysis

T-Test

ANOVA

Ordinary Least Squares Regression

5. Exploratory Data Analysis

The average listing gain across all IPOs was 4.29%.

The average subscription level was 25.94 times.

Among 264 IPOs:

31 Blockbuster Success IPOs

47 Strong Success IPOs

67 Moderate Success IPOs

68 Loss-Making IPOs

51 Disaster IPOs

The analysis also revealed that IPO performance improved significantly after 2017, with 2020–2021 showing the highest average listing gains.

6. Demand Analysis

Subscription demand emerged as the strongest predictor of IPO success.

Average listing gains by subscription category:

A T-test comparing high-demand and low-demand IPOs produced:

T-statistic = 6.93

p-value < 0.001

The results indicate a statistically significant difference in listing performance.

7. Institutional Participation

Very high institutional interest generated average listing gains of approximately 28%.

Institution-driven IPOs produced average gains of 9.66%, while retail-driven IPOs generated negative average returns.

However, regression analysis suggests that the independent impact of institutional participation weakens after controlling for subscription demand.

8. Issue Size Analysis

ANOVA testing was performed to analyze the effect of issue size.

Results:

F-statistic = 5.09

p-value = 0.006

Medium-sized IPOs generated the highest average listing gains, while small IPOs produced negative average returns.

Issue size influences IPO performance, although the relationship is less consistent than subscription demand.

9. Market Sentiment Analysis

NIFTY 90-day returns were used as a proxy for market sentiment.

Correlation with listing gains:

$r = 0.23$

Regression analysis showed a weak positive relationship between market sentiment and IPO performance.

Market conditions support IPO success but do not independently drive listing gains.

10. Regression Analysis

Multiple linear regression was conducted using:

Log Total Subscription

Log Issue Size

Institutional Participation

Market Sentiment

Results:

$R^2 \approx 0.22$

Subscription demand remained highly significant.

Issue size showed moderate significance.

Market sentiment showed marginal significance.

Institutional participation lost significance after controlling for demand.

11. Key Findings

Subscription demand is the strongest predictor of IPO success.

High-demand IPOs significantly outperform low-demand IPOs.

Institutional participation improves outcomes but largely through demand.

Market sentiment has a weak positive influence.

Issue size exhibits mixed effects.

12. Conclusion

This study demonstrates that investor demand is the most consistent and statistically significant determinant of IPO listing performance in India.

While institutional participation and favorable market conditions contribute positively, their effects are secondary compared to subscription demand.

The findings provide useful insights for investors, analysts, and market participants seeking to evaluate IPO opportunities.

13. Tools and Technologies

Python

Pandas

NumPy

SciPy

Statsmodels

Excel

Power BI

14. Dashboard

An interactive Power BI dashboard was developed to visualize:

IPO market overview

Demand analysis

Institutional participation

Final research findings

The dashboard enables users to explore the key drivers of IPO success through interactive visualizations.